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Measurement and Definition of Gentrification in Urban Studies and Planning

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Abstract

The amount of empirical research on the extent, causes, and consequences of gentrification continues to expand. This article reviews the methods utilized to delimit and quantify gentrification in the literature. Such measurement is undertaken in order to assess the consequences of gentrification, such as displacement, over various time scales and geographies. Recent research has demonstrated that the use of different quantitative definitions of gentrification may yield conflicting research results, and further, can muddy the waters for policymaking and political discourse. This article critically assesses the breadth of quantitative definitions and offers recommendations for future studies.

Keywords

gentrification, neighborhood change, displacement, literature review, measurement

Introduction

During the first two decades of the twenty-first century, gentrification appeared ascendant in American cities. The phenomenon, described as "in-migration of middle- and upper-income households into existing lower income urban neighborhoods," became a focus of attention in policy and scholarly debate (Griffith 1996, 241). Despite the global recession from 2007 to 2009, gentrification persisted and arguably accelerated in the last decade. Hwang and Lin found that in US cities "since the 1970s and especially since 2000, downtown gentrification has strengthened and a growing number of neighborhoods have gentrified" (2016, 12). In response to this escalation, conversations about the issue among the press, politicians, and the public increased in tenor. Scholars reacted and have attempted to assess the impact of gentrification on cities and their residents in the current era through qualitative and quantitative studies (Brown-Saracino 2017).

Gentrification is a critical issue for city planners because it shifts new residents and financial capital into disinvested lower income areas, which are typically home to incumbent low-income and racial minority populations. Thus, gentrification is a directly observable contest between the haves and have nots in urban space, a conflict spotlighted by scholarly and public attention to economic and racial structural inequality particularly as it plays out in the housing market. Further, the social problem of displacement, potentially caused by gentrification, is also concerning to planners and policymakers (Marcuse 2015). Displacement is the forced removal or blocked relocation of residents out of or into certain areas that have experienced rent or home price increases and further includes the phenomenon of indirect displacement via social and cultural shifts (Slater 2009). Debate about displacement is inseparable from social and economic inequality in cities today, especially as it intersects with race and culture (Hyra 2017).

Some scholars argue that despite "becoming more prevalent in US cities… rigorous research on the extent, causes, and consequences of gentrification remains rare" (Ellen and Ding 2016). Ellen and Ding's qualifiers *rare* and *rigorous* notwithstanding, this review will show that the literature on gentrification and its consequences have expanded greatly. This literature can be roughly separated into two groups, qualitative and quantitative, with the former generally relying on analysis of several or individual neighborhoods selected through a priori knowledge of gentrification, and the latter typically relying on census or other data to conduct panel or cross-sectional analysis of gentrification in multiple areas (Brown-Saracino 2017).

Scholars who choose not to define gentrified areas through a priori or purely qualitative means must first determine if neighborhoods are eligible to gentrify through some quantitative metric and then specify criteria by which neighborhoods did in fact gentrify over some time periods (Galster and Peacock 1986). Van Criekingen and Decroly (2003) noted that across the literature, there was no unanimously approved empirical delimitation of the concept of gentrification forty years after Glass (1964) coined the term. Barton (2016) agreed and further noted that "the strategy used to identify gentrified neighbourhoods potentially

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had important implications for whether gentrification was associated with other neighborhood outcomes" (2016, 2).

Drew completed a review of differing strategies and noted that "differences in the data and variables used to separate gentrified from non-gentrified neighborhoods… can lead to very different conclusions about where gentrification occurs, as well as the outcomes with which it is associated" (2018, 1). Easton et al. (2020) further elaborated upon this point by noting that accurate identification of gentrifying neighborhoods is critical in research measuring displacement, because neighborhoods not marked as gentrifying are used as baselines for comparison. Despite this point, Easton et al. note that "operationalisation of these dimensions [of gentrification] in terms of measurable variables is far more equivocal [than agreement on gentrification's broad meaning]" (2020, 288). Both Preis et al. (2020) and Mujahid et al. (2019) compared different gentrification measurement methods and reached similar conclusions, with Preis et al. concluding that "cities that adopt one of the various different mapping methods will come to very different conclusions about the location and severity of gentrification based on the method they choose" (Preis et al. (2020), 16).

These papers, however, have been limited in scope. Drew (2018) summarizes six quantitative methods and discusses several more, Preis et al. (2020) compares four, Mujahid et al. (2019) compare three, and Barton (2016) compares two. In aggregate, these reviews have not assessed more than a hundred extant additional quantitative methods used to define gentrification and measure its consequences. In this article, I expand the scope of such reviews by connecting the choices on methods scholars have made, using a novel descriptive analysis technique to conduct categorization of a much broader set of papers, to theory on the causes of gentrification. The aim of this review is to provide both a comparison of methods used to quantify gentrification, grounded in theory, and also provide suggestions on methods that future scholars seeking to measure gentrification can use to inform their studies.

The review finds that across a set of nearly two hundred papers that utilize quantitative methods, there are tendencies towards measurement of the same time periods using limited time intervals. Further, measurement is more commonly limited to a single city, rather than multiple cities or regions. Quantitative definitions of gentrification tend to be simple, and authors often utilize basic metrics rather than complex analysis tools to identify it. Further still, these metrics are more often than not vaguely defined. Robust analysis which combines census data with field surveys remains rare despite evidence that such approaches add nuance (Hammel and Wyly 1996).

The structure of this review is as follows. First, I review definitions of gentrification and theories about how and why it is caused. I next review the effects of gentrification. Following that, I review the general methods with which scholars have identified gentrification. I then review my approach for identifying the variety of methods that have been used to empirically delimit gentrification in quantitative urban studies research. The next section offers a description of the results of the classification process. In the penultimate section, I discuss the results of the analysis and relate those results to trends in gentrification research. In the final section, I offer conclusions and suggestions for improvement of further quantitative research on gentrification and its consequences.

Defining Gentrification, Understanding Its History, and Reviewing Its Causes

Definitions and History

"Gentrification" was introduced as a term by Ruth Glass, a British sociologist, in the early 1960s when an urban gentry was beginning to reinhabit and rehabilitate the housing stock in working class neighborhoods in London. In her analysis, Glass (1964) described how these affluent newcomers transformed neighborhoods from renter occupied to owner occupied, increased property prices, and displaced working class residents. Glass described gentrification as "an inevitable development, in view of the demographic, economic, and political pressures to which London, especially Central London, has been subjected," arguing that gentrification is a neighborhood process that cannot be attributed to one factor, but is instead the result of a set of metropolitan interactions (Glass 1964, xviii).

Glass's terminology began to enter the lexicon in the United States in the early 1970s. Perhaps best encapsulated by the *brownstoning* movement in the Park Slope neighborhood (purchase and rehabilitation of older, large townhomes) in Brooklyn, New York; the press and the general public became aware of the process of gentrification, alternatively termed the "back to the city movement," or "central city revival" at the time. Across the country, in large cities like Washington D.C., Philadelphia, and Chicago, professionals and bohemians alike were defying the conventional habit of seeking more land and larger homes in the suburbs. They instead sought to rehabilitate and inhabit what they deemed as valuable prewar housing stock in neighborhoods close to downtowns (Lipton 1977).

Though early research generally focused on gentrification as an isolated process in residential housing markets in certain cities, it became clear during the 1980s and even more so after that this process was widespread, and part of a broader cycle of economic and class change in cities across the world. Initial research that speculated the process was transient was proven wrong over time through identification of the process in additional cities into the 1990s. Recent macroscale studies have found that the pace of gentrification has since accelerated (Hwang and Lin 2016).

Scholars in the current era, and I in this article, generally agree that gentrification is defined by an influx of new investment and new residents with higher incomes and educational attainment into a neighborhood (Chapple and Loukaitou-Sideris 2019). In the United States, gentrification is often associated with the process of racial demographic change in inner cities. In the prototypical and widely observed case, mostly White higher social class individuals move in to formerly disinvested areas that are largely Black or Latinx. Despite that association, scholars have argued that gentrifiers can be of any race as social class is not hegemonic within or across racial groups, and indeed some have found limited evidence of Black and Latinx gentrification (Hyra 2006; Moore 2009). Because of this nuance, race is not typically used as a binary or deciding factor in defining gentrification.

Critical scholarship on the left has identified and emphasized how the traditional description of gentrification as in-movement of higher income residents leaves out a story of power imbalances, marginalization, and alienation drawn along racial lines, especially in the United States (Betancur 2002). Communities with little access to capital or political power, particularly urban Black and Latinx communities, have little ability to influence the planning process or organize to block private investment that may cause displacement or other unwanted impacts. Stated directly, some argue that structural racism, which excludes low-income minority groups from planning and economic power, has left open the ability for the combined forces of mostly White private investment and political power to gentrify inner cities via in-movement by White residents (Smith 1996). Recent empirical research has added nuance to the story, finding that gentrification in some cities is more likely in diverse and immigrant neighborhoods and less likely in deeply segregated areas (Hwang 2015). Debate over the scale and impacts of displacement is ongoing and contentious (Brown-Saracino 2017).

Gentrification has attracted the attention of city residents, decrying housing cost increases and displacement; popular culture, following the trend and tastes of millennials moving into central cities; the press, noting the "rebirth" of central cities in the aftermath of the greatest housing crisis in generations and the plight of residents subject to higher rents; and the political class, responding to these provocations. Increasing societal attention paid to income inequality and structural racism have pushed the issue even deeper into the national consciousness. In order to evaluate the consequences of the phenomenon to inform policy, scholars have built on work that began in the 1960s that identified where, why, and how the process occurred.

Why and How Does Gentrification Occur?

As evidence of gentrification accumulated in the 1960s and 1970s across major cities in North America, scholars began to attempt to understand the root causes of the process. Publications on the causes of gentrification in the first decades of gentrification research can be simply classified into two groups: the demand side, which sought to understand the demographic and social forces that influenced the desire for gentrified housing and in-movement to disinvested areas, and the supply side, which sought to understand the economic conditions which created neighborhoods with housing stock that could be gentrified (Griffith 1995). In more recent decades, scholars have moved beyond the supply and demand debate and shifted their focus from attempting to answer questions of causality toward examination of consequences (Lees, Slater and Wyly 2008).

Before recounting the supply–demand debate, it is worth noting that gentrification sits nested within work, which seeks to understand neighborhood change. Gentrification is a specifically defined upward transition in social class in a neighborhood, but many have studied the variegated forms of neighborhood social and economic change, which include stasis, upgrading, and decline. Prominent theories of neighborhood change include the filtering model, the bid/rent model, the tipping model, and the border model (Temkin and Rohe 1996). The filtering model, first advanced by Hoyt (1933), describes how newly built housing eventually degrades in quality due to decreased investment, thus allowing lower income residents to replace more affluent residents as those residents seek newer, higher quality housing. Gentrification may represent part of a filtering process, but instead of new housing being constructed at the suburban fringe, rehabilitated or new housing is constructed in denser, urban areas. Such a process may be reflected in the bid/rent model, commonly referenced in urban economics, and advanced by Muth (1969). The bid/rent model describes the tradeoffs between leisure time, housing cost, and commuting cost; as desire for lower commuting cost increases due to an increasing value of time, some higher income consumers may choose older housing near jobs in the city rather than larger housing in the suburbs, thus gentrifying inner areas (Laska and Spain 1979). Border and tipping models examine the impacts of social class and race on in-and-out movement from neighborhoods within the framework of the invasion-succession model, assessing how trends in racial change may cause neighborhoods to "tip" (Schwirian 1983). All of these models frame debate over gentrification, which is one specific type of neighborhood change. Early debate on gentrification, however, was to some degree separate from this context and instead focused on competing supply and demand side explanations.

Work on the supply side owes its foundations to Neil Smith, a geographer and student of noted urban theorist David Harvey. In his seminal article "Toward a Theory of Gentrification: a back to the city movement of capital, not people," Smith laid out his argument, founded in Marxist economic theory, that over time, land parcels in central cities saw reduction in their "capitalized ground rent" relative to their "potential ground rent" (Smith 1979). As structures aged and became less valuable, the capitalized ground rent of the land would fall, and landlords would receive less income. While this occurs, the potential ground rent, or the rent that could be achieved through utilizing the parcel at its highest and best use, may increase over time for certain parcels as the city's economy grows. Enterprising capitalists would eventually identify the rent gap between the capitalized and potential ground rents and purchase and rehabilitate structures to be used in some updated manner to thus earn the full potential rent (Smith 1979).

Smith's (1979) initial work was in response to early research on gentrification in the 1970s that had sought to explain the process of gentrification from the perspective of the gentrifiers, or the demand side. Scholars sought to understand who gentrifiers were, why they chose gentrifying neighborhoods, and the larger structural factors that influenced their decisions. Ley (1986) and Hamnett and Williams (1980) were both influential researchers in this area, as they sought to identify the characteristics of the demand that sustained the production of the supply side. Their and other work linked the decline of the industrial economy and rise of the service sector to gentrification, observing that as professional employment increased in central cities, the residential neighborhoods nearby became more desirable, especially as undesirable industrial uses vacated (London, Lee and Lipton 1986).

The forces of supply and demand which have interacted to drive decades of gentrification did not do so in a free-market vacuum. In many cases, government—at the local, state, and federal level—has been directly or indirectly involved in the process of urban regeneration or gentrification. As Griffith (1995, 246) noted, "local governments throughout the United States have been among the most active participants in the gentrification process." The public sector can catalyze or accelerate gentrification through changes to zoning and form-based codes, investment in physical infrastructure like transit, and the provision of services (Zuk et al. 2018). Governments can also provide financial incentives to businesses and developers to locate in certain gentrifying neighborhoods or even provide financial incentives to gentrifiers themselves through instruments like property tax abatements. Others have linked federal housing policy to gentrification efforts, particularly the Housing Opportunities for People Everywhere (HOPE VI) program which sought to demolish and redevelop subsidized housing and thereby deconcentrate poverty and create mixed-income neighborhoods as a replacement (Goetz 2011). Critical research has focused on the impact of increased legal restrictions on homelessness and beautification and intense regulation of public spaces, all of which can make cities more amenable to the preferences of gentrifiers (Mitchell 2003).

This combined set of forces—political, supply, demand—acts across the market economy for land in the city and region. Some scholars have sought to separate these forces and identify causal factors that catalyze or influence gentrification in specific neighborhoods or parts of neighborhoods within cities or regions, and they have found that supply, demand, and political factors are related to the process of gentrification. In a review, Brown-Saracino (2017) found that at the neighborhood level, these factors include locational attributes, increasing gentrification's likeliness in neighborhoods near cultural amenities, a downtown center, public transportation, and other gentrifying places; there are also more hyperlocal factors like the quality of the housing stock, presence of single family homes, and the age of buildings. Hwang and Sampson (2014) found that racial demographics matter, as certain neighborhoods who are majorly Black are less likely to gentrify than more diverse places, a finding replicated by Timberlake and Johns-Wolfe (2017). Others have emphasized the spatial dependence of gentrification—the fact that places adjacent to wealthy or already gentrified areas are often the next to gentrify or experience redevelopment (Meligrana and Skaburskis 2005).

Research has shown that the process of gentrification occurs due to a complex, interacting set of forces that apply at different scales. At the regional scale, gentrification is the result of a complex set of demand side and supply side factors influenced by a legacy of political and economic exclusion and public sector decisions. At the neighborhood scale, the process results from a complex set of geographic and socioeconomic factors relating to the built environment, city amenities, and the residents who live in those places. The consequences of gentrification, however, are a subject of great debate. There is a divide between the macroscale quantitative body of work that finds gentrification to be of muted impact and a microscale qualitative body of work that finds it to be of high consequence to neighborhood residents. Brown-Saracino (2017) identifies a core source of this issue: methodological differences in measurement of gentrification, which consist of limits of data across time and space and disagreement over how to qualify gentrification itself with various variables or techniques. This article offers a full inventory of those measurement issues, a task as yet incomplete and of importance as the empirical literature on consequences continues to expand. Before reviewing that inventory, the next section reviews research on the consequences of gentrification and the recent responses of planners and the public sector to those consequences.

The Effects of Gentrification

Gentrification occupies a unique position in the urban policy debate. Unlike clear social ills like concentrated poverty or environmental injustice, gentrification has boosters and is often an indirect policy goal of local government. Few elected officials or planners in communities in the United States categorically oppose in-movement of capital or higher income residents into lower income areas that they represent. In fact, many elected officials have touted gentrification, as they stood to benefit through expansion of the tax base and therefore potential mitigation of urban problems (Beauregard 1985). Planners and elected officials seek economic improvement in the lives of their constituents and understand that in-movement of capital and new residents could bring that about. These individuals with planning power must hold that desire while responding to residents' concern about gentrification, especially in regard to displacement and escalating housing costs. This contradiction and conflict is deeply rooted in the intra- and intermetropolitan competition for growth, jobs, and prestige. In planning departments, this conflict is especially sour. Planners attempting to advance the tenets of smart growth, designed to advance economic and environmental sustainability, must "strengthen and direct development towards existing communities" (Smart Growth Network n.d.). Some argue that in doing so, "urbanists have prescribed compact development without evaluating the very real consequences of new, dense construction in terms of raising land prices beyond the means of current residents" (Chapple and Loukaitou-Sideris 2019, 3). This debate is a direct manifestation of the conflict between the “just city” and the “growing city” in Campbell (2007) planning triangle.

Brown-Saracino’s (2017, 517) review of scholarship on gentrification noted it is increasingly viewed in qualitative and microlevel studies as a social issue that is “deeply problematic and consequential for longtime residents.” That is only half the debate. In the same review, Brown-Saracino (2017, 520) notes that macrolevel quantitative analyses offer a viewpoint that gentrification is not as widespread as commonly thought and further and more importantly argues that “displacement is far from endemic.” These quotes reflect a deeply embedded debate on gentrification and displacement in urban studies, regarding whether or not the two are inseparable. Chapple and Loukaitou-Sideris (2019, 40) argue that “displacement occurs when forces outside people’s control force them to move from their residence. Because these forces may stem from either disinvestment or investment, displacement is not necessarily directly induced by gentrification.” Slater (2009) argues that gentrification was always linked to inequality and class struggle and was thus inseparable from social justice and displacement.

A significant fraction of the empirical work on the consequences of gentrification is focused on the question of displacement, or the forced or involuntary removal of incumbent residents in gentrifying neighborhoods due to increased housing costs, or blocked relocation of low-income residents into gentrified neighborhoods (Slater 2009). The incumbent residents studied in this work, who are to be replaced or joined by gentrifiers, are typically of lower social status and earn lower incomes and are often racial minorities. A growing body of macroscale evidence indicates that low-income residents of gentrifying neighborhoods do not face higher displacement rates than other residents in high-income neighborhoods or low-income neighborhoods that are not or cannot be gentrifying, even after controlling for numerous factors (Freeman 2005; Ellen and O’Regan 2011; Ding, Hwang and Divringi 2016; Delmelle and Nilsson 2020). Some more nuanced evidence on that question has shown that displaced residents are more likely to relocate to lower income areas and that homeowners are less likely to be displaced than renters (Newman and Wyly 2006; Ding, Hwang and Divring 2016). Qualitative evidence has shown that at the local level, long-term residents are displaced and often negatively psychologically impacted during gentrification processes (Pattillo 2007; Betancur 2011; Hyra 2017).

The consequences of gentrification stem far beyond displacement and can be both positive and negative. Atkinson (2004) lists and reviews positive and negative consequences of gentrification. The positives include stabilization of declining areas, increased property values, decreased vacancy rates, increased government revenue, encouragement of further development, reduction of sprawl, and increased social and income mix. Negatives include displacement, community conflict, loss of affordable housing, increased homelessness, industrial and commercial displacement, loss of social diversity, and cultural displacement (Hyra 2017). Freeman (2006) via in-depth qualitative work found highly nuanced perspectives, both positive and negative, from incumbent residents on gentrification in their gentrifying neighborhoods in New York City. Some residents may appreciate greater access to services and stores and benefit from increased home prices if they are property owners; others resent feeling unwelcome or are unable to afford new amenities designed for the newcomers. Incumbent residents may clash with newcomers over differences in cultural and social expectations. Atkinson (2004) also notes that research on the benefits of gentrification is much rarer than research on the negative impacts, but this is a point that requires further explanation. Scholars who have sought to explain or invent the benefits of gentrification have often used other words—urban renaissance, revitalization, or regeneration. These scholars may avoid use of the word gentrification, which is a universally recognized label for urban inequality (Slater 2006).

Lang (1986) found limited evidence in Philadelphia that gentrification produced increased revenue streams for local government, and further this effect was greater than the cost of providing additional amenities for these residents. Other older research was inconclusive on positive impacts of gentrification and instead emphasized that attention paid to real or potential positive impacts was magnified by the context of urban decline prior to the 1990s (Beauregard 1986).

The positive and negative impacts of gentrification may impact the life course of residents who remain or are displaced, through increased or decreased access to education and jobs, shifts in public health, exposure to crime, or other means. A burgeoning body of research has attempted to assess the financial and health impacts of gentrification on residents who remain or are displaced from gentrifying neighborhoods. This research was partly spurred by Mindy Fullilove’s thesis that residential displacement, caused by urban renewal or other factors, causes traumatic psychological stress (Fullilove 2016). In a review of studies that relate gentrification to health impacts, Schnake-Mahl et al. (2020) find mixed results, which vary by outcomes assessed, measurement methodology, and local context. Several studies have investigated the link between falling crime and gentrification, finding that less crime in central city neighborhoods makes them more attractive to potential gentrifiers (Ellen, Horn and Reed 2019). Other empirical studies have assessed the impact of gentrification on crime in gentrifying neighborhoods, generally finding a negative relationship after an initial increase due to destabilization of social control (Kirk and Laub 2010). Kreager, Lyons and Hays (2011) find a curvilinear relationship between gentrification and crime, suggesting early stage gentrification is associated with small increases but later stage gentrification associated with crime declines. That relationship is true only for property crime, not violent crime, which is unassociated with gentrification. Papachristos et al. (2011) find that gentrification’s effect on crime is contingent on race, with White gentrifying neighborhoods seeing declining homicide rates and Black neighborhoods seeing increasing robberies. Barton (2016) found in New York City that sub-boroughs gentrifying at a faster clip experienced much larger declines in serious crimes than other sub-buroughs.

A small body of work has identified a process of industrial change and labor market transition which impacts local residents and workers in gentrified areas. Curran (2007) and Lester and Hartley (2014) identify a shift from blue collar industrial employment to service sector employment in gentrifying neighborhoods, with the latter authors finding evidence of more rapid employment growth. Meltzer and Ghorbani (2017) build on these results and find that incumbent residents experience job losses in local gentrifying areas, but those losses are balanced by increases in employment in surrounding areas. Limited evidence shows that wages are higher for residents who remain in gentrifying neighborhoods (Ellen and O'Regan 2011). Hwang and Lin (2016) find positive relationships between gentrification and credit scores for those who stay versus those who exit gentrifying areas.

A review of the methodological breadth of scholarly work that identifies the consequences of gentrification is outside of the scope of this article, and it is of note that no such review yet exists. This is partially due to the explained cross-disciplinary nature of the gentrification literature, in which scholars from planning, criminology, sociology, economics, public health, and additional disciplines have attempted to test the impact of gentrification on a wide variety of social metrics. Typically, these studies will identify gentrifying neighborhoods—with methods explained in detail in the next section—and test, with qualitative or quantitative methods, whether those neighborhoods see different social outcomes than neighborhoods that do not gentrify. Authors generally hypothesize that gentrification itself will lead to measurably different social dynamics in gentrifying neighborhoods. The classic example of this is the measurement of residential displacement: do low-income households move out, or get evicted at, higher rates in gentrifying neighborhoods than comparable households in nongentrifying neighborhoods? Such questions may be answered with macrolevel data in the aggregate, with average rates derived from census or other data, or in qualitative work, through interviews. Others may rely on individual-level panel data to track the behavior or characteristics of individuals over time as they enter or exit gentrifying neighborhoods. Many papers identify change by testing it over time, meaning they compare change in a base year against a final year for the studied metric in both gentrifying and nongentrifying areas. Econometric methods utilizing regressions with varying levels of complexity have become commonplace as a method to test hypotheses. If one replaces the example of displacement with typically studied metrics in other disciplines, like crime rates, travel behavior, spending patterns, social capital, or network structure, etc., then one can infer the scope of papers that have been produced. All of these papers rely upon identification of gentrification at some geographic level, contingent on available data for the consequence being measured, to test such hypotheses. In the next section, I review the methods that authors have used to identify gentrifying places.

Measurement of Gentrification

For decades, scholars have used qualitative and quantitative metrics to identify gentrification, track it over time, and measure the consequences of the process in gentrifying areas. A few key parts of the identification and measurement process can be identified by referencing the origins of the term. Ruth Glass noted that gentrification occurred in disinvested, central areas of London that were largely working class. These areas over time experienced both an influx of financial capital—discussed in her work as investment in existing housing stock—and an influx of new residents of a higher social class. Empirical studies of gentrification there and elsewhere have followed this framework by first identifying gentrifiable areas, which meet some criteria for disinvestment and/or lower social class occupation, and then by specifying which of those neighborhoods gentrify over some time period (Galster and Peacock 1986). Those undertaking this measurement process must therefore account for time, a unit of spatial analysis, a set of criteria for eligibility, and a set of criteria that qualifies as gentrification. The most common approach is to follow that framework and use of quantitative census data at some small level of geographic reference that approximates the neighborhood—for example, the census tract or block group—and assess socioeconomic and demographic conditions at separate census intervals (Barton 2016).

Time

Within the standard approach, scholars typically assess conditions at decadal census intervals, though five-year intervals and multidecadal intervals of twenty and even forty or fifty years have been utilized where and when possible. Such intervals are most often used because they are the time periods at which socioeconomic and demographic data for small geographic units are made available through Census results. This includes two major drawbacks: many years are missing direct observation and the decadal observation points do not align with the macroeconomic business cycle. Despite this, scholars have found some justification in the lengthy interval due to the fact that neighborhoods change slowly in their social and economic composition due to the relatively static nature of the housing stock (Zwiers 2018). Others have argued for the importance of studying gentrification over multiple decades (Meltzer 2016). Since 2009, researchers have gained access to the US Census Bureau's American Community Survey data, which provides five-year data averages for small levels of geography, released annually. Such data enables, for example, scholars to compare the year 2000 with the 2005–2009 period or any other subsequent five-year period after those years (though temporally overlapping time periods should not be compared).

Spatial Unit and Analysis Area

Within both the qualitative and quantitative branches, scholars in most cases focus on a particular area of study, or a particular unit of analysis, for a broader investigation of multiple places. Those relying on census data are in the United States and elsewhere confined to using units of statistical aggregation offered to them, which are generally census tracts. In the United States, tracts are considered an imperfect proxy for neighborhoods but have long been used in studies of neighborhood change and gentrification (Timberlake and Johns-Wolfe 2017). Beyond tracts, most researchers tend to use government-defined definitions when studying neighborhood change, primarily for reasons of convenience related to data availability (Kirk and Laub 2010). A census tract is a small subdivision of a county, home to an average of 4,000 residents, and it is made up of smaller block group units. These subdivisions of counties are created to be homogeneous relative to demographics and socioeconomics, which may potentially bias estimates of neighborhood change as their boundaries are not random with respect to socioeconomic conditions (US Bureau of the Census 1994). Census tract boundaries can and do change over time, so researchers frequently make use of spatially harmonized tract boundary datasets from private or academic sources to ensure consistency.

With the geographic unit of analysis selected, researchers must also select a reference area in which to conduct their analysis. Some papers consider entire or even multiple, metropolitan areas; some individual cities or sets of cities; while some consider certain neighborhoods within cities or even single neighborhoods. Still others consider rural areas specifically (Nelson, Oberg and Nelson 2010). Many have limited their eligible areas to "central cities," often defined with political boundaries or with quantitative metrics like proximity to a central business district, population isochrones, or areas with older housing stock (Freeman 2005; Hwang and Lin 2016). Researchers here are trying to hold the original conception of gentrification as an inner-city or inner-area phenomenon affecting older neighborhoods. This is not a universal trend, however, as many have included suburbs (Ley 1986).

Variables and Criteria for Analysis

With time, a spatial unit, an analysis area, and variables chosen, researchers must mark areas as eligible to gentrify and assess whether they did or did not. It is worth noting that in some work, scholars opt not to limit the set of tracts, and all tracts are considered eligible to gentrify. Researchers often use a threshold to mark tracts as eligible, holding tracts below or above some level of a quantitative metric, like median income, as eligible. For example, Freeman (2005) marked tracts as eligible if they had a level of housing construction below a twenty-year metropolitan median and a median income below the metropolitan average.

With a set of tracts or block groups defined as eligible, researchers must then determine whether tracts gentrified or not. Within this part of the measurement process, there is a much wider range of methods. For example, Yonto and Thill (2020) measured change in a social status index composed of three variables (occupation, education, and median household income). Tracts with above mean increases in the index were categorized as having experienced gentrification. Others have taken a simpler route, and like with the eligibility threshold, created a gentrification threshold. For example, Freeman (2005) utilized a five-step process to identify gentrification: to qualify as eligible, tracts must be located in the central city, have a median income less than the metropolitan 40th percentile, have a smaller proportion of housing built in the last twenty years than the metropolitan 40th percentile; and to gentrify, tracts must see a greater increase in tract educational attainment than the increase in median metropolitan area and see an increase in real housing prices over the measurement period.

Other studies branch out from census data to assess whether or not tracts gentrified or mark them as eligible for gentrification. Mixed methods studies have utilized field surveys of buildings over time to assess physical upgrading, coupling that information with census data to assess change over time (Hammel and Wyly 1996; Wyly and Hammel 1998). In a unique stream of work, Smith, Duncan and Reid (1989) and Smith (1999) utilized tax arrears data to identify the point of temporal shift between disinvestment (unpaid taxes) and reinvestment, a trackable indicator of supply-side pressure toward gentrification. More recent work has used video or computer imagery, from services like Google's street view, to allow large-scale field surveys of building and neighborhood conditions to be compared against census data (Hwang and Sampson 2014). Still others have assessed gentrification with parcel-level data on building values, renovations, and sales (Helms 2003). In some quantitative work, scholars have referenced city plans, newspaper articles, visual assessments of structures, and the perceptions of residents (Brown-Saracino 2017). This branch of research typically uses a priori definitions of gentrified areas, assuming that certain areas are gentrifying or have gentrified, coupled with resident surveys or interviews to assess perceptions and consequences of the process.

Review of Quantitative Methods Used to Define Gentrification

A search for "gentrification" in the title of publications in Google Scholar returns over 7,900 results, with numerous duplications and frequent instances of work from nonpeer-reviewed sources (e.g., think tanks, city governments, news media, or advocacy organizations). To create a more useful database, I used the software *Harzing's Publish or Perish,* a publication database search tool which pulls articles based on author or title or other searches, to scrape Google Scholar with the search term gentrification applied to article titles (Harzing, 2007). The top 980 unique results ranked by citation number were pulled, ranging in publication date from 1973 to 2020.

To further limit the set for analysis, books and book chapters were removed, along with prepublication working papers, papers published at think tanks and other nonpeer-reviewed venues, and work deemed extraneous to urban studies. Next, 35 papers published in languages other than English in non-English language journals were removed. Finally, papers published outside of the US-UK-Canada-Anglo-Oceania context were removed. Although papers from outside this context may be relevant to gentrification theory and provide important empirical evidence, these studies are about urban regions subject to very dissimilar urbanization dynamics and planning regimes.

In this next stage, I divided those papers into two camps: qualitative, in which gentrification is defined and studied without quantitative means, and quantitative, in which gentrification is measured through census or survey data or other numerical means. In order to divide the papers, I read the abstracts and the data and methods sections. The end goal was to obtain a complete set of papers in which a quantitative metric was used to define that at least one place is gentrifying or gentrified. Those two options, however, left out a wide body of theoretical, criticism, and commentary pieces, so those papers were moved into a third category. The flow chart in Figure 1 illustrates the entire classification process. After a manual review of the 169 quantitative papers, ten additional important citations were added in. These were missed by the initial search process and obtained from reference lists in the set of 169 papers.

The quantitative papers were classified using a matrix in an Excel spreadsheet, manually, with information supplied by reading the methodology or other relevant sections. I gathered information to fill the matrix based on the following questions.

- What years were used as the start and end period of the analysis, and what was the corresponding range?
- What was the geographic unit of analysis and geographic scope of analysis?
- What was the primary data source?
- What demographic or socioeconomic variables were used to (i) mark areas as eligible to gentrify and (ii) measure change over time that indicates gentrification?
    - Were these variables representative of the demand side or supply side theories on the causes of gentrification?
- Was an index or other multicomponent computational factor like principal components analysis used to measure gentrification?
    - Was some absolute numerical threshold used to mark neighborhoods as eligible or as having gentrified?

Results

Time and Geography

Researchers have chosen a wide variety of time periods to conduct their analyses of gentrification. Figure 2 is a scatterplot showing the start year of analysis on the x-axis and the end year on the y-axis. The figure also contains a table at bottom right listing the number of papers which start or end their analysis in each decade. The scatterplot illustrates that there is a cluster of papers at the top right; these are papers that begin analysis after 1990 and end before the year 2010. This is visible in the corresponding table, which shows sixty-six of 179 papers beginning in the 1991–2010 period (37%) and ninety-five ending in that period (53%). Another visible trend is a fair number of long-range analyses beginning in the 1960s or 1970s and not ending until the year 2000 or later (the top central part of the chart). No studies in the dataset ended earlier than the 1970s, and only thirty-five (20%) began prior to that date.

Figure 2 is further illuminated by data on interperiod measurement range. For example, an author may have studied 1970–2010 but within that period measured gentrification from 1970 to 1990 and 1990 to 2010. The vast majority (65%) of researchers measure gentrification over a timespan of approximately a decade. Only thirty-seven papers (21%) measure demographic and/or economic change over a greater than eleven-year period to declare whether or not gentrification has occurred in a given place. Forty-two papers (23%) measure changes over relatively short time less than a decade scales or assess data at single point in time.

Turning toward geography, we observe further evidence of the relatively consolidated nature of research on gentrification. Nearly half of the dataset (85 papers, 47%) limits analysis to a single city only, and a sizeable minority (15%) of papers are even further limited to one neighborhood or several neighborhoods in a single city. The remaining papers, just over one third of the set, tackle analysis at a larger scale. These papers pool data across multiple cities, an entire region or multiple regions, or even in a few cases, an entire country.

A majority of papers (52%) have relied on the census tract, or its equivalent in other countries, to assess gentrification. Only a minority of papers (14%) relied on units of geographic aggregation smaller than the census tract, including a few that rely on parcel-level data. Nearly a third of papers, at 27%, used analysis areas that are larger than census tracts but still subdivisions of cities or towns. These subcity level areas include various other census aggregations like zip codes or transportation analysis zones and political subdivisions of cities like wards or boroughs. The remaining papers assess gentrification at larger scales: entire cities, entire counties, or at self-defined geographies such as preselected areas of countries or counties or villages.

Data Sources and Variables Used for Definition and Assessment

The source of summary data used in research about gentrification is deterministic to the extent that scholars must rely on public data sources like the census, which produces data at fixed intervals and fixed geographic scales. Census data at narrow levels of geography are typically only available in the postwar period, further limiting the potential for analysis prior to the 1950s. Nearly three of four papers have used census data in some way. The next largest share, only 14%, have

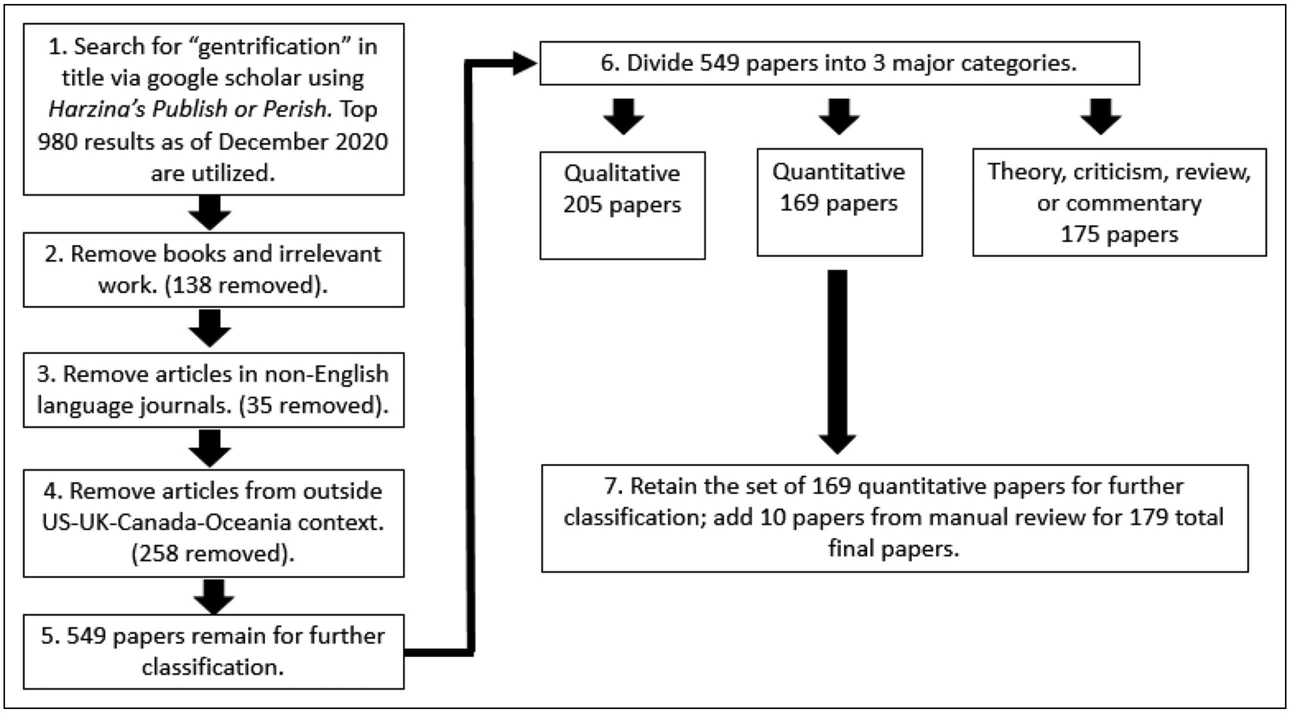

Figure 1. Flow chart to select papers for analysis.

relied on a noncensus government source such as local government data on property sales to study neighborhood change. Prior to the availability of extensive, analysis-ready census data, some authors relied on private market demographic surveys in the 1970s and 1980s. For example, Henig (1980) utilized data from the now defunct R. L. Polk & Company's *Profiles of Change*. A group of papers has utilized field or personal surveys to assess change over time as well.

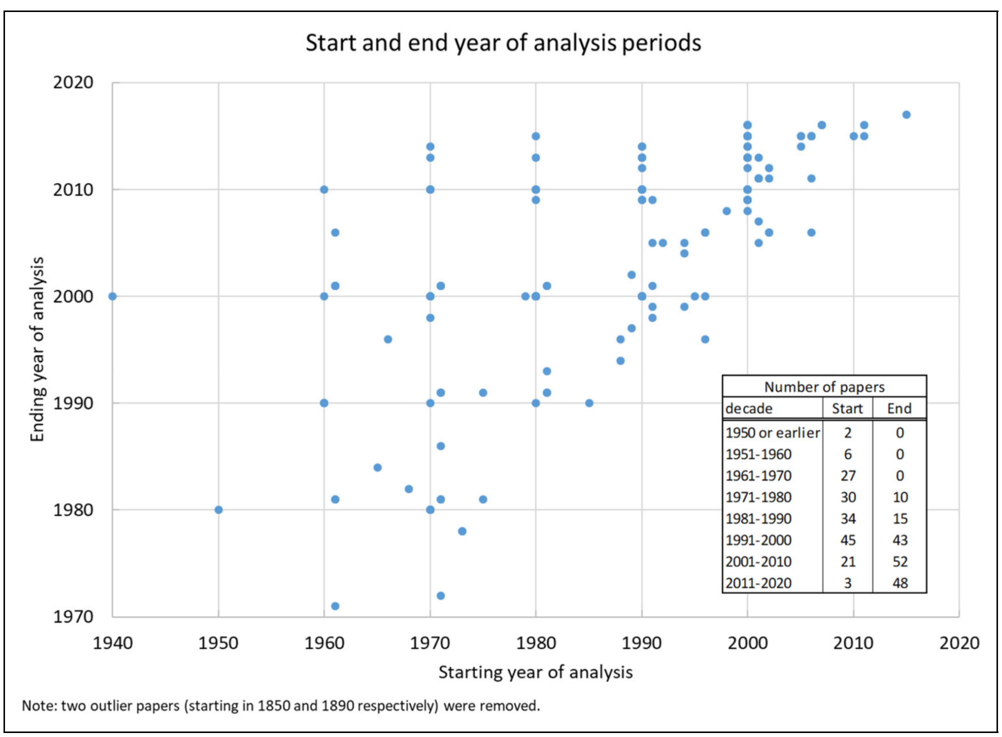

Figure 2. Start and end year of analysis scatterplot.

With a data source chosen, researchers must pick certain metrics or variables from that data source to study gentrification. These metrics and variables can be classified according to gentrification theory, which explains the causes of the phenomenon can be divided into two categories: supply and demand. Table 1 presents a list of variables used in each category.

On the demand side, authors have utilized demographic indicators to describe the populations who are moving in to gentrified areas: age, ethnicity, family composition, foreign born share, household size, population change or density, and race. Socioeconomic indicators include educational attainment, income, mobility rates, occupation, poverty rates, housing cost burden, and shares receiving public assistance. Much of this data is available in decadal or more frequent census surveys like the American Communities Survey; however, publicly available versions of this data do not track the same individuals over time, such that different time periods of data form a time series of the then-current residents of the neighborhood. Other data sources like the Panel Study of Income Dynamics do track the same families or individuals over time with great detail on their income and location, but the sample size is greatly limited which makes assessing aggregate neighborhood effects more challenging. Other indicators include the locations of coffee shops, judged by several authors to be proxy for demand from certain groups (Hwang and Sampson 2014). On the supply side, authors have used a variety of metrics to assess the composition and characteristics of the built environment. Typically, these indicators focus on the housing stock: property age, sales data, home values, rent prices, recent construction, unit and building characteristics, tenure,[1] and vacancy rates. Other supply side variables include the share of units used for short-term rentals, green roofed structures, data on loans in neighborhoods, property tax amounts and rates, data on tax arrears, and visual surveys of the quality or characteristics of the built environment.

Nearly half of the papers (45%) use both supply- and demand-side variables to track change. A large plurality (29%) utilizes only demand-side metrics, and a much smaller share (7%) use only supply-side metrics. Note that thirty-three papers (18%) used an a priori definition of gentrified areas and are classified as such.

Table 1. Variables Used to Identify or Study Gentrification.

Demand	Supply
Age of residents	Airbnb rental share
Number and location of coffee shops	Number of green-roofed structures
Educational attainment of residents	Home prices (average, median, or from sales data)
Employment or unemployment rates of residents	Housing age
Ethnicity of residents	Housing stock change
Family composition (share of couples or childless couples, share of same sex couples)	Loan data
Family or household income (average or median)	Number of bedrooms in housing units
Foreign born share	Property taxes
Gini coefficient	Rent price (median, gross, average)
Household size	Sales data
In-movement/mobility rates	Structure type
Occupation	Tax arrears data
Per capita income or median income (individual)	Tenure
Population change	Unit size
Population density	Unit type
Poverty	Vacancy
Race	Visual surveys
Rent burden (share paying over certain income)	
Share of artists	
Share receiving public assistance	

Methods of Defining or Measuring Gentrification

The methods used to measure whether or not gentrification occurred exhibit the most variance. As such these methods are hard to classify, but some trends are evident. The simplest definitions declare some increase in a single variable in a study area, like educational attainment or home prices, to be gentrification. Complexity increases from there. Nearly a quarter of the papers, forty-one (23%), in the set utilized statistical techniques to create indices or more precisely determine variables which indicate the presence of gentrification. One method is to use a weighted or nonweighted index of a set of demand- or supply-side variables. For example, Ley (1986) equally weights the share of population in white-collar jobs and the share of population with a university education in census tracts to create a social status index. More complex methods utilized include principal component analysis of differences over time, a statistical technique which reduces a set of variables to provide a smaller set of the most influential (Bereitschaft 2020). These more complex techniques allow authors to input a large number of potential causal factors and withdraw the most influential, rather than preselecting a few commonly utilized variables like education, occupation, or home prices.

The majority of the papers in the dataset, however, do not use such a method to combine variables. These seventy-seven percent of papers track changes in individual variables, sometimes multiple variables, over time to indicate the presence of gentrification. Eighty-two papers (46%) used some discrete threshold to determine whether gentrification occurred in a specific place. Freeman (2005) influenced numerous future papers by creating a threshold of comparison to change in the metropolitan area, without using an index or more complicated scheme. This original work marked eligible census tracts as gentrified when they experienced an increase in educational attainment greater than the metropolitan area average increase and also an increase in real housing prices. Others take a different approach and use increases in variables or indices relative to other tracts. For example, Hwang and Lin (2016) constructed a socioeconomic status index from educational attainment and household income data at the tract level and used that index to rank tracts in a metropolitan area. Census tracts which experienced a two-quartile increase in the socioeconomic status index index over any decade from 1960 to 2010 were said to gentrify. This differs, for example, from the approach of Ley (1986), which simply took the top twenty percent of tracts in the index, rather than tracts that passed some threshold increase.

Patterns and Correlations in Quantitative Research Methods

The structure of the data gathered for this section enables comparison of certain trends in quantitative methods. For example, it is possible to ascertain if there are significant differences in methods between papers that attempt to study gentrification at the city or subcity level versus papers that study gentrification across cities, a region, or multiple regions. Such comparisons can point researchers to methodological gaps in certain areas of study. A complete breakdown of these results is outside of the scope of this article due to the possible number of permutations. For example, one could also attempt to identify methodological differences between papers that focus on supply- or demand-side variables or ascertain methodological differences between papers that study short- or long-time scales or tease out the differences between papers that use different geographic units of analysis, etc. However, some important patterns are visible in the data, which are noted here

First, as aforementioned, papers in about one third of the dataset study gentrification in multiple cities, a region, or an entire region. The remaining papers limit their geographic area of analysis to a single city or portions of a single city. While there were no discernable differences in time period or geographic unit of analysis between these groups, the larger scale papers tended to use more complex methods. 30.0% of the larger scale papers used an index or other multivariate statistical operationalization of gentrification, but only 18.5% of the smaller scale papers did. The larger scale papers also were much more likely to use a specific threshold to qualify gentrification: 57.6% versus 39.0%. The larger scale papers were also much more likely to utilize both supply and demand metrics: 62.1% against 36.3%. Less than a fifth of papers in each group included race as a metric, with no discernable difference between the two. These results imply that papers analyzing gentrification patterns or consequences at a larger scale tend to use more statistical rigor and also consider gentrification more holistically than smaller scale papers.

The dataset also reveals that complexity in gentrification research has been increasing recently. Papers using an index, principal components analysis or other statistical operationalization of gentrification were published with an average date of 2013, versus 2008 for those without such methods. These papers also averaged using a longer time period (13.7 years) than those without such methods (10.2 years), and used more recent data, with an average end year of 2006 versus 2002. 58.5% of the more statistically complex papers used both supply and demand metrics in the definition of gentrification, against only 42.0% of the less complex work. The more complex papers were also more likely to use a larger geographic study area, with 48.7% using an area larger than one city, against 33.0% for the less complex papers. The next section compares these and other trends and draws conclusions based on the theoretical foundations of gentrification.

Discussion

A number of conclusions can be drawn from the trends presented in the previous section. Beginning with the data on time, the two figures taken together illustrate that the majority of what is known about gentrification has been determined using data from 1980 to 2010. While some work has examined earlier periods, the 1950–1970s remain understudied. Some of this is due to a desire to respect Glass' discovery of the phenomenon in London in the 1960s, as most research has examined periods after that date to see if the process progressed or became evident elsewhere; it is also an issue of data availability. Gentrification theory makes no requisite of time period and if demographic and socioeconomic data can be coupled with data on home or rent prices or other supply side indicators prior to the 1960s, then further research would be welcome.

Continuing with a critique on time, a general consensus exists to measure gentrification as occurring over ten or fewer years within that time range. However, across the literature, there is little to no theoretical discussion of why a ten-year interval, against a twenty-year or even thirty-year interval, is the proper time length of measurement. It is apparent that the choice of time interval is made based on data constraints, but limiting it to a single decade on average is arbitrary. A stream of the literature discussing stage models of gentrification has found that the process follows a geographically idiosyncratic but still relatively predictable time path. Condensed, this argument states that gentrification does not happen all at once, and certain things happen first: slow in-movement of artists or lesbian, gay, bixexual, transgender, or queer individuals, with modest changes to tenure rates and rents and slower change to demographics and socioeconomics; then upgrades to the housing stock and in-movement of higher status groups and more rapid change; then new construction, displacement, commercial gentrification, and even super-gentrification (Kerstein 1990; Lees 2003; Van Criekingen and Decroly 2003). Little research has been done to measure the lengths of time it takes for these processes to occur.

As with the data on time periods, the data on geography shows a lack of variance in gentrification research in terms of geographic scale. The majority of papers look only at single cities or subdivisions thereof and study change over time using a neighborhood analogue like the census tract. It is also worth noting that over half the dataset has investigated gentrification in a single city or parts of a city; large-scale comparative research across cities or metropolitan areas remains rare. The question for researchers on geography then is to identify if gentrification is unique within cities or even neighborhoods or if the same trends and patterns occur in different cities over time.

Most scholars have understood Glass' original conception of gentrification as an inner-city phenomenon, yet some scholars have over time studied at the scale of entire cities, metropolitan areas, suburbs alone or multiple suburbs, and even rural areas. This geographic extension of research has been fruitful as authors have found evidence of gentrification in suburbs, small towns, villages, and even rural areas (Brown-Saracino 2017). Some analysts have criticized these approaches, such as Bondi (1999) who asked if it was time to put the term gentrification to rest or Maloutas (2011) who argued that spatiotemporal expansion of the definition of gentrification introduced a reduction in theoretical rigor. Others, such as Slater (2006) disagreed, arguing that it made no sense to focus Glass's fine empirical geography given dramatic change to global and local economies, cultures, and labor markets since the 1960s. Despite these arguments, authors who do undertake large-scale studies of metropolitan areas or regions treat tracts in central cities the same as tracts elsewhere, meaning these tracts must pass no different "test" to be marked as gentrified based on their location. Socioeconomic or demographic change outside cities may, as Slater suggests, be of a different character than in central cities meriting divisions of these geographies in analyses. This potential area of research remains poorly explored, as most quantitative studies which include suburban or rural areas treat those areas no differently than central areas.

Regarding geographic unit, authors of these papers have employed a wide variety of scales to measure analysis but tend to stick with the best neighborhood analogue, the census tract. Quantitative work at scales smaller than the census tract remains rare, and it is here perhaps that the most improvement could be offered. Analyses of gentrification at the tract level are limited to what census averages can describe about a population, masking detail on the distribution of income, educational credentials, and rents or home prices. In the United States, census tracts can be relatively spatially large and have economically and demographically dissimilar pockets of neighborhoods within them. Without further geographic detail, these details are impossible to discern. A few scholars have improved upon this process by using block group or parcel-level data (Helms 2003; Lee and Newman 2021).

Turning toward data sources used in the analysis, it is tougher to critique the choices made by authors. All are greatly limited by data availability and must rely on the census for demographic and socioeconomic information. That data can in some cases be coupled with local data on housing supply and characteristics from other government sources. Hammel and Wyly (1996) created a national-level resource in the United States with their field survey of neighborhoods which can supplement census data; their data has been used by numerous successive papers (Freeman 2009; Moore 2009; Hwang 2015, 2016). The time commitment and resources required for this approach, however, have meant that it has not been replicated. Several have utilized Google Street View to undertake a similar field survey of structures (Hwang and Sampson 2014; Ilic, Sawada and Zarzelli 2019). Overall, however, there has been limited effort to cross-fertilize aggregate census averages with more specific local-level data in the manner argued for by Hammel and Wyly decades ago.

Eighty-two papers (45.8%) have included explanatory variables or definitions of gentrification that include both supply- and demand-side items. That said, a plurality of papers (52, 29.0%) utilize only demand-side variables in order to understand the characteristics of gentrifying or displaced populations. These papers leave out measurement of home or rent prices, a puzzling choice given Glass's inclusion of investment in her definition. Metrics like occupation, education, and home or rent prices remain the most commonly utilized. Most papers which included a demand side metric (a total of 134) included either education or occupation or both as a specific variable. A total of 115 did so, or 85.8% of papers measuring the demand side (64.2% of the entire dataset).

Race remains an uncommonly used metric, as only thirty-one papers (17.3%) included it in the operationalization of gentrification (note that a larger share of papers may have studied impacts or consequences with respect to race race). This is likely due to Glass's theoretical foundation of gentrification as contingent on change in neighborhood social class composition, which is commonly measured with average or aggregate occupation or education statistics. In the United States, race and social class correlate, but the recent findings on the nuanced links between race and gentrification, especially from Timberlake and Johns-Wolfe (2017) and Hwang (2015, 2014), clearly indicate that race and social class should both be addressed in empirical work.

Within the set of thirty-one papers that include race as a metric, all but two of the papers were published after the year 2000, and twenty were published after 2016, reflecting increased attention paid to the links between gentrification and race. Most of these papers (20) measured gentrification with both supply and demand variables, and all but five also included education or occupation as a metric (and all five of those also utilized income or poverty, both proxies for social class). The papers exhibited no clear trends in whether they measured gentrification across a single city, region, or multiple cities or regions, but most (22) utilized census tracts as their unit of analysis.

On measurement technique, the data revealed that analysis of gentrification still relies on relatively simple metrics. Only a quarter of papers studying gentrification with quantitative metrics have relied on more complex instruments like indices or factor analysis to define the process as occurring; most rely on simple increases in a single variable or multiple variables. Beyond that, the metrics used to qualify places as gentrifying tend toward vagueness. Less than half of the work assessed used some sort of specific threshold to mark whether or not gentrification occurred; the remainder used unspecified criteria to declare areas as gentrified or not. Such imprecision muddies the waters for discourse on gentrification and analysis of its consequences. Despite that, some patterns emerged from the data which showed that methodological complexity is increasing over time. Newer papers tend to use more complex methods and study gentrification at larger areas, while also addressing both supply and demand. These papers also utilize longer time scales and more recent data. This positive trend of increasing methodological nuance is undoubtedly crucial for deriving more meaningful conclusions about the consequences of gentrification.

Conclusions

Over forty years of study of gentrification with quantitative methods has produced a wide range of evidence on its presence and continued expansion throughout cities across the developed world. This expansion has been linked causally to both positive and negative consequences, though little research has assessed the positive and only recently has research turned to assess negative consequences other than displacement. Significant debate is ongoing as to the depth, prevalence, and severity of residential and commercial displacement of incumbent residents and businesses during the process of gentrification.

This article revealed a number of trends through a systematic review of 179 papers published since the 1970s that sought to measure gentrification and its consequences. Authors have tended to measure gentrification over the same time periods and at the same geographic scales, due to data availability limitations. Analysis of time intervals is dominated by analysis at the decade time scale, despite little theoretical discussion of why or how that amount of time is the proper measurement. Geographic units of analysis are dominated by tracts or larger scales, with few papers analyzing gentrification at the block or address level. Definitions of gentrification are typically simple, and use of complex quantitative techniques to identify the process remains rare; further, definitions are often vague or arbitrary. These findings empirically validate the conclusions of Easton et al. (2020), wherein the authors noted that identification of gentrified neighborhoods is contested in research, as gentrification occurs unevenly across time and space, and spatial units used in research are too large to identify consequences like displacement that occur at the local level.

Gentrification research has emphasized the complexity of the process. No one definition is gospel. Future research on gentrification should consider utilizing multiple definitions of gentrification to assess consequences. This article also highlighted that few papers have mixed methods, in which authors join census or other data with personal or field surveys to ground-truth gentrification processes; this has continued despite the influence of a series of papers in the 1990s by Daniel Hammel and Elvin Wyly. Authors should also ponder the findings of this paper and consider grounding their choices of measurement interval and geographic unit in theory before relying on decadal averages or census tracts as sole metrics of analysis. Further, despite the phenomenon's links to racial inequality in the United States, few papers have considered race as a component in defining gentrification.

One major limitation of this work is that it did not classify the qualitative empirical literature, which is an even larger group (205 papers compared with 179 quantitative) in the dataset of 980 papers (Figure 1). While assessing the methodologies of the qualitative literature was outside of the scope of this review, scholars seeking to empirically examine the consequences of gentrification should familiarize themselves with the qualitative literature. Brown-Saracino (2017) provides an indispensable overview and summary of contrasts between measurement techniques and methodological foundations in the qualitative and quantitative literatures. Qualitative scholars focus much less on identification gentrification's breadth and scale and instead focus on questions about the characteristics of gentrification on the ground, through analysis of the choices and beliefs of participants and bystanders in the process (Brown-Saracino 2017). Barton (2016) finds that quantitative work may illuminate gentrification in areas where qualitative work has paid less attention, due to less media or social visibility in certain neighborhoods. Qualitative work, however, can answer questions on the real impacts and perspectives of incumbent residents and gentrifiers in a way that quantitative work cannot (Freeman 2006). Both Barton (2016) and Brown-Saracino (2017) suggest that each side of the methodological divide can and should innovate through attention to the methodologies of the other.

Planners and researchers seeking to identify gentrification in their communities should take several lessons to heart from this review. First, use of neighborhood- or subneighborhood-level demographic and economic data is recommended, as scholars have identified that neighborhood-level factors influence gentrification, and this data is now easily accessible. Second, if possible, it is recommended that scholars pair socioeconomic and demographic data with physical surveys of neighborhood conditions or the perceptions of residents. Third, effort should be made to pair metrics that track both the supply and demand sides of the gentrification equation. A novel approach would be to combine time—matching associated economic cycles in metropolitan housing supply and demand—with data on both the supply and demand side. Further regarding time, this study has made clear scholars and planners must reasonably ground their measurement range and periods based on knowledge of local conditions.

While the coronavirus disease 2019 pandemic has introduced uncertainty into the global economy and spawned a rash of opinions about the future of cities, scholars of gentrification know that the phenomenon has outlived dramatic changes to the global economy before. The 2020s will undoubtedly yield additional quantitative work on the subject as gentrification mutates and continues to spread.

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Supplemental material

Supplemental material for this article is available online.

Notes

1. Tenure could be considered both a supply or demand indicator, but here is classified as a supply indicator as it illustrates the potential for conversion from renter occupancy to owner occupancy.

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